Experiment-8: Lambda Function with DynamoDB and API Gateway

Aim:

The primary goal of this experiment is to design and implement a serverless architecture using **AWS Lambda**, **Amazon DynamoDB**, and **API Gateway**. Specifically, the objective is to create a Lambda function that can perform **CRUD operations** (Create, Read, Update, Delete) on a DynamoDB table and expose this backend functionality through an API Gateway endpoint.

Procedure:

Before going through the steps Login to your **aws** free-tier account.

Step-1:

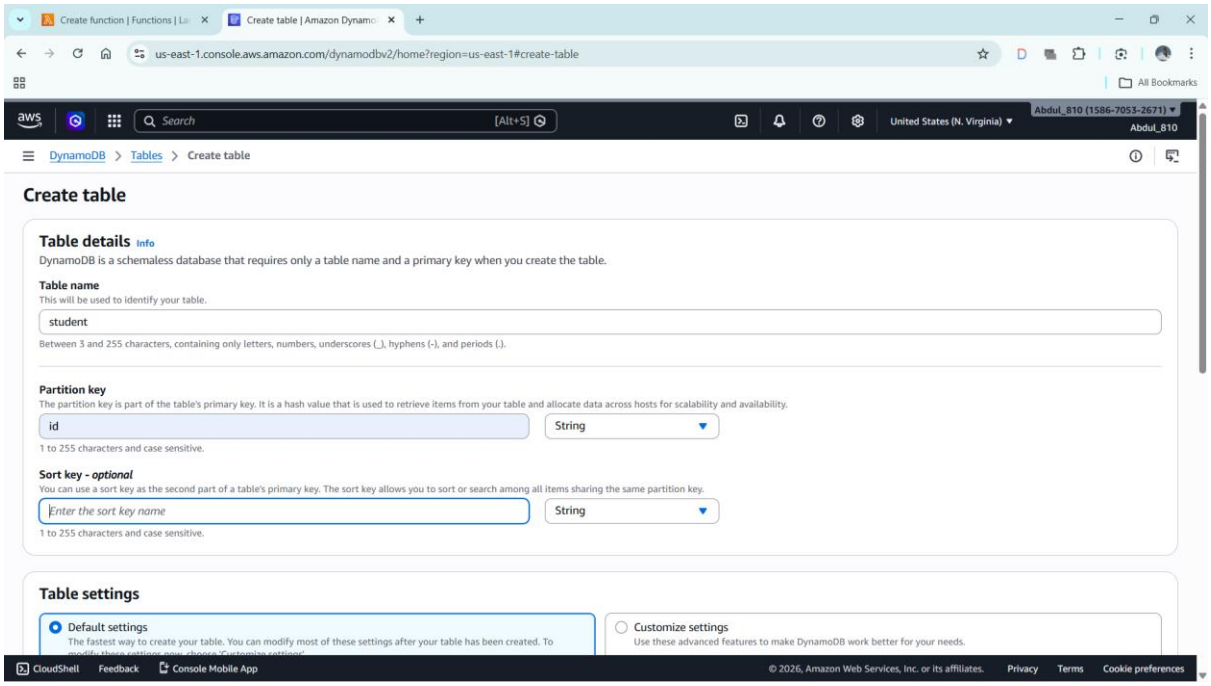
Create a DynamoDB table with the following options:-

Table Name: student

Partition Key: id

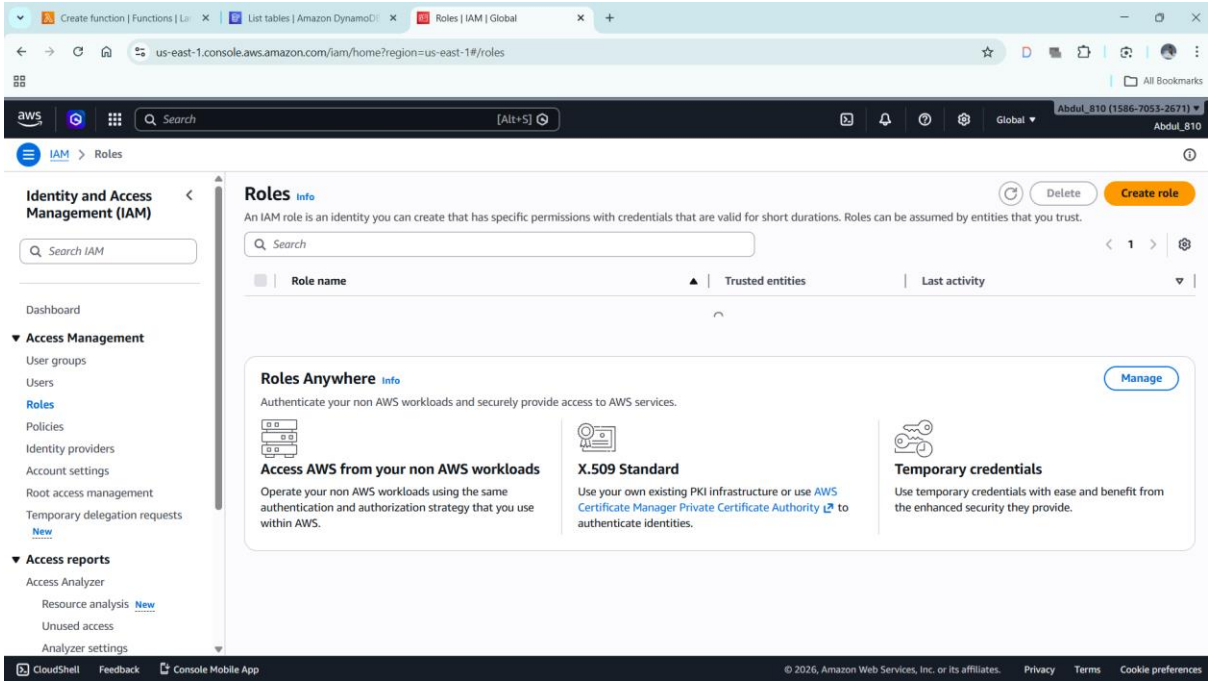
Data Type: String

Only make changes for the above three fields for creating a table as shown below:

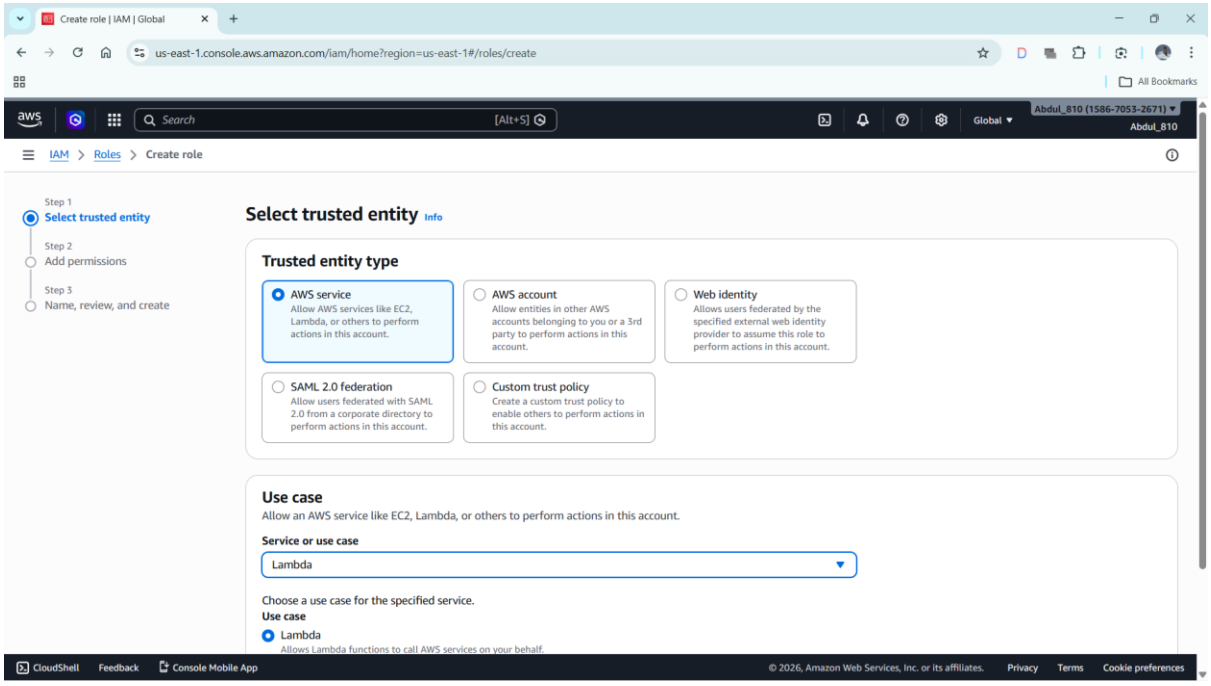


Step-2:

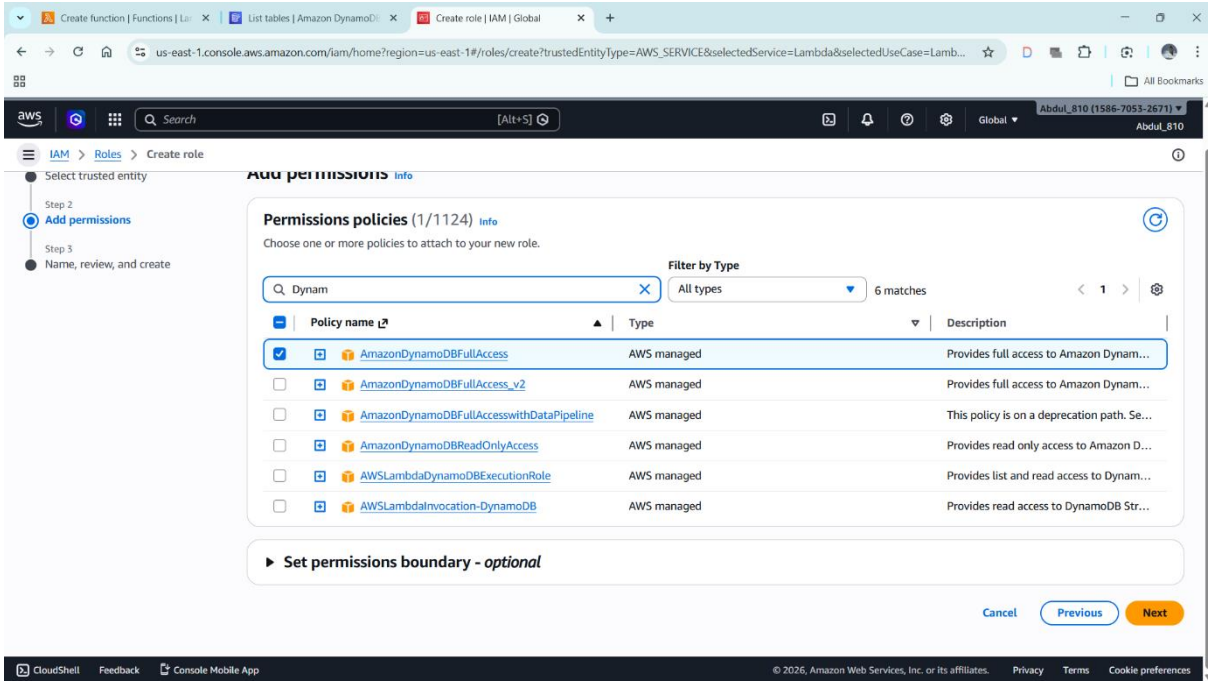
Later, search for the IAM service and select the service, go to roles and click on “create role”.



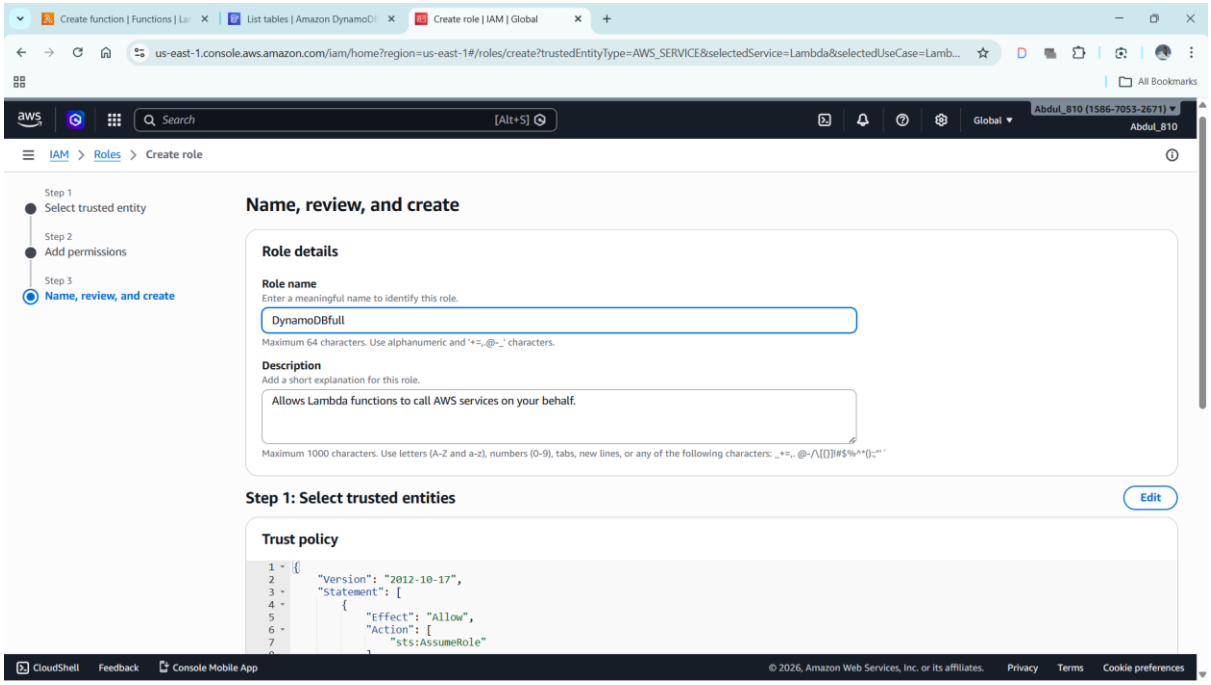
Then, select trusted entity type as “AWS service” and the use case as “Lambda” and click on next.



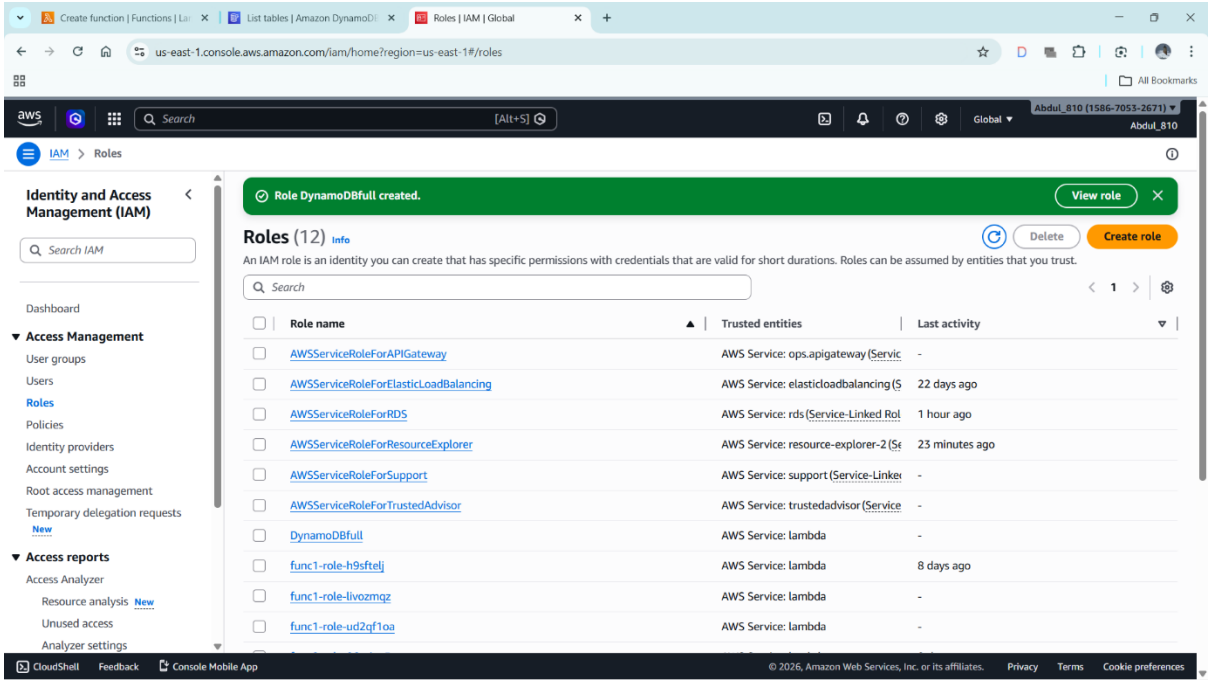
And then, attach the policy named- “AmazonDynamoDBFullAccess” and click on next.



Now, provide the role name as “DynamoDBFull” (any other, if you want) and click on create.



And finally, you should be able to see the created role in the list successfully.

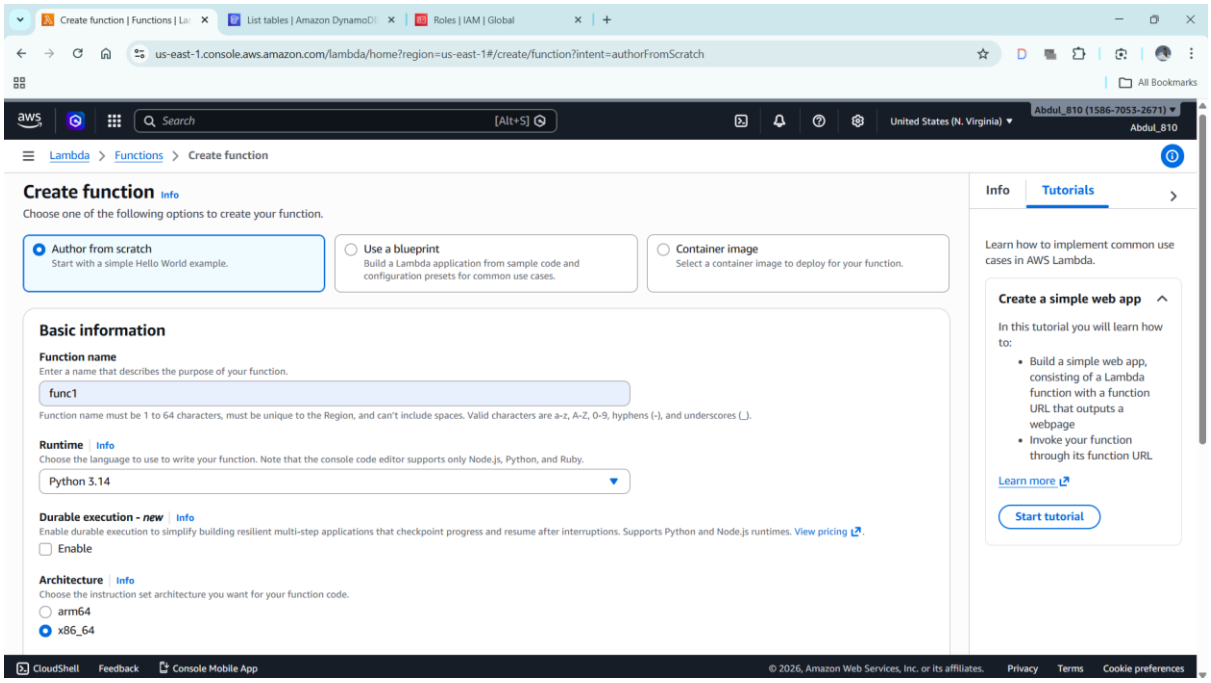


Step-3:

Search for the lambda function and select it and click on create a function.

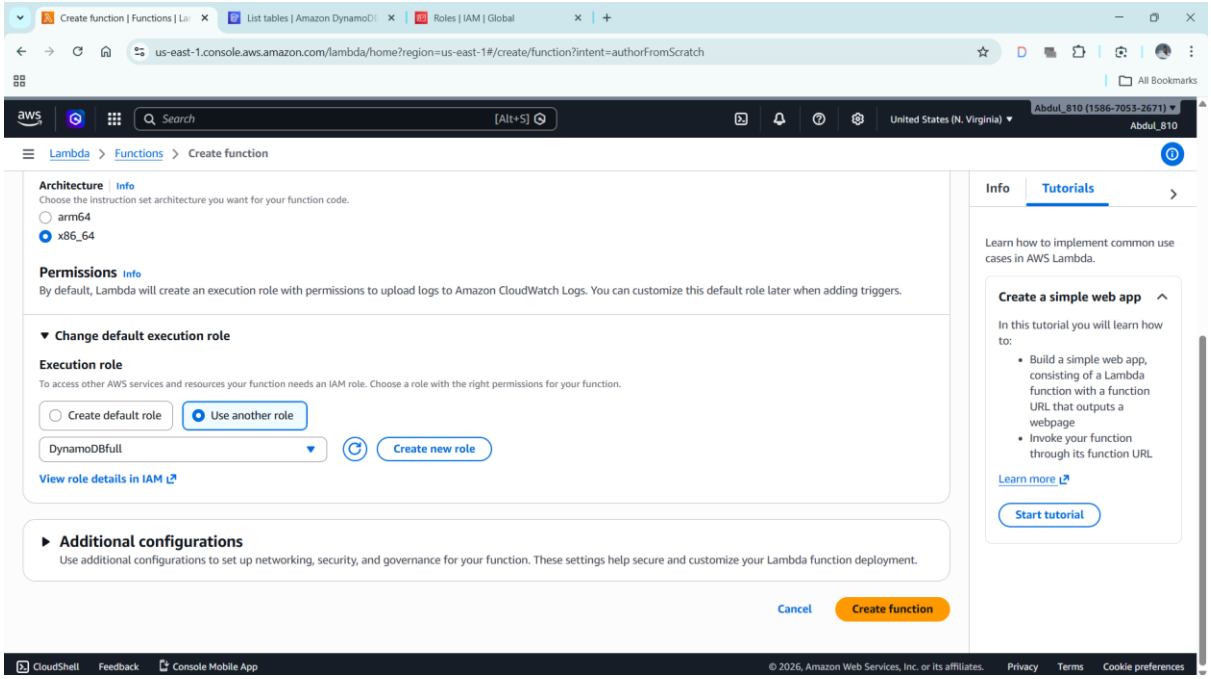
Name: func1

Runtime: 3.14(Python)

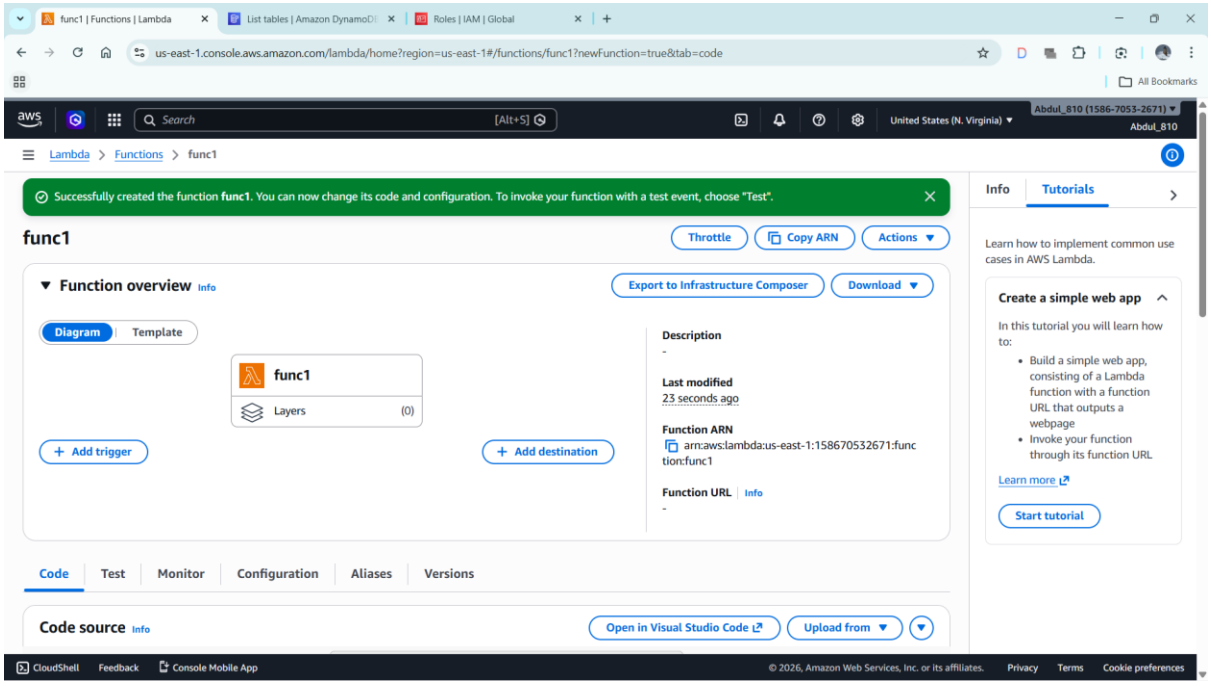


Change default execution role as: use another role and choose the role which you have created earlier with name “DynamoDBFull” from the dropdown menu.

And then click on create a function.



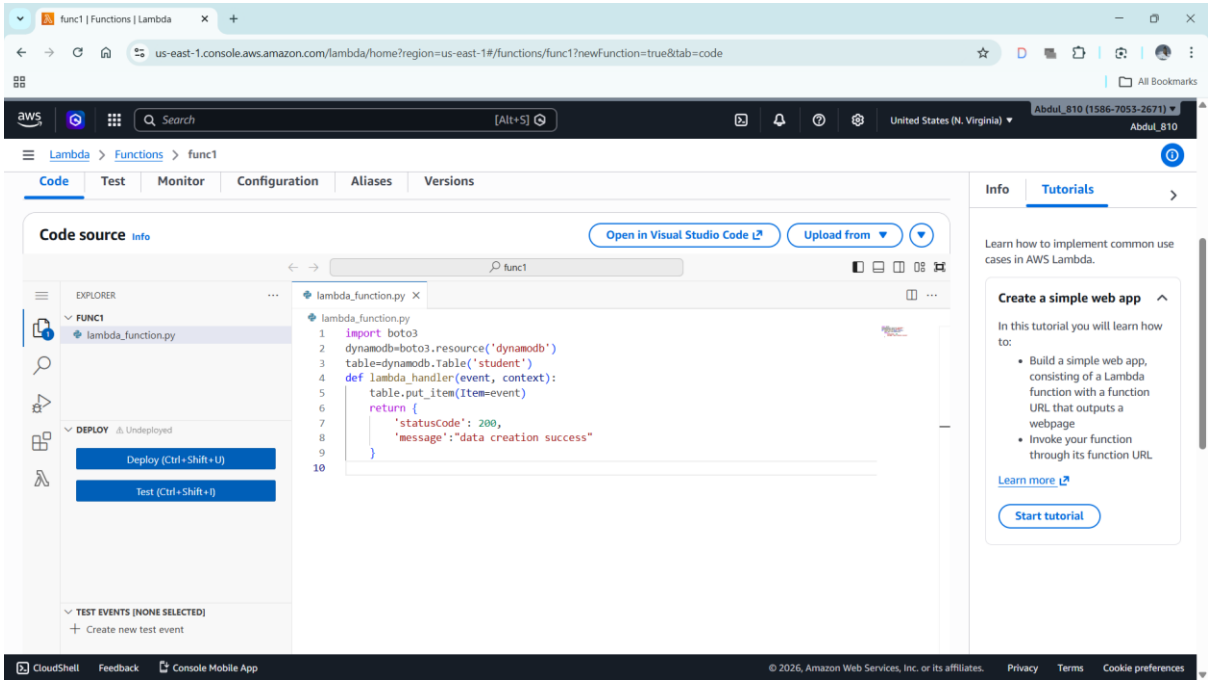
You should be able to see the newly created function:



Step-4:

Go to the code tab in the function available and you will find a python editor over there: Add the sample code in lamda_function.py file.

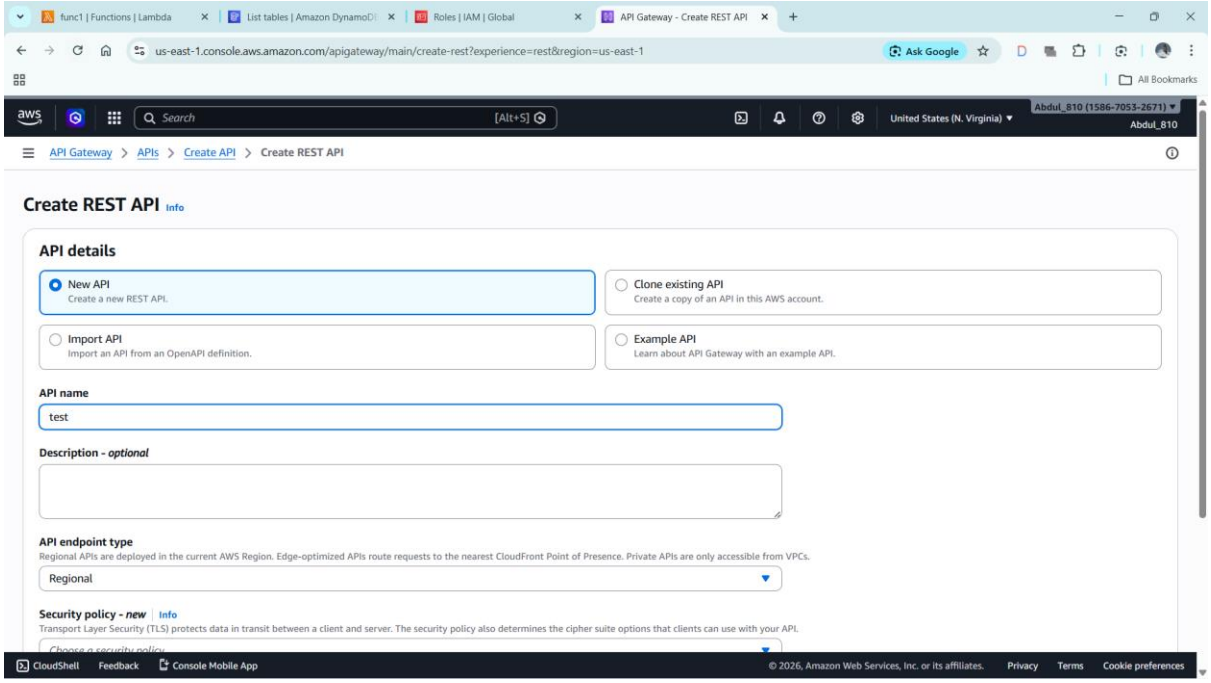
```
import boto3
dynamodb=boto3.resource('dynamodb')
table=dynamodb.Table('student')
def lambda_handler(event, context):
    table.put_item(Item=event)
    return {
        'statusCode': 200,
        'message':"data creation success"
    }
```



And then click on “Deploy”

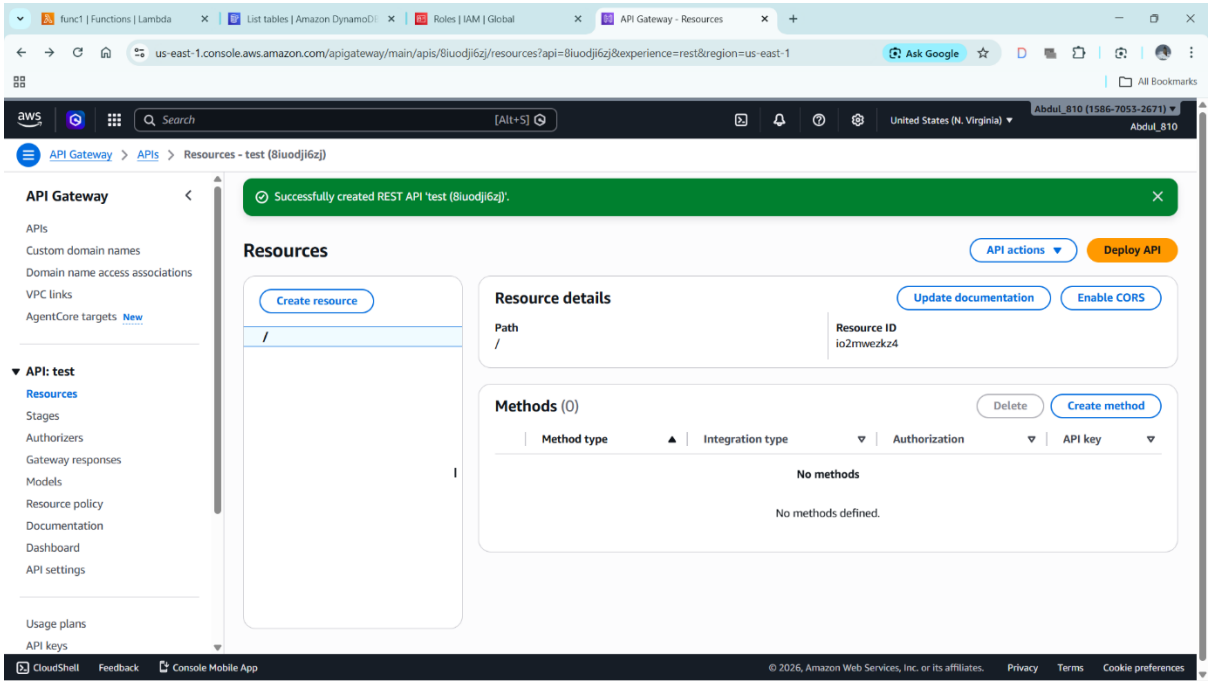
Step-5:

Search for API Gateway service and select it and click on create an API and choose REST API

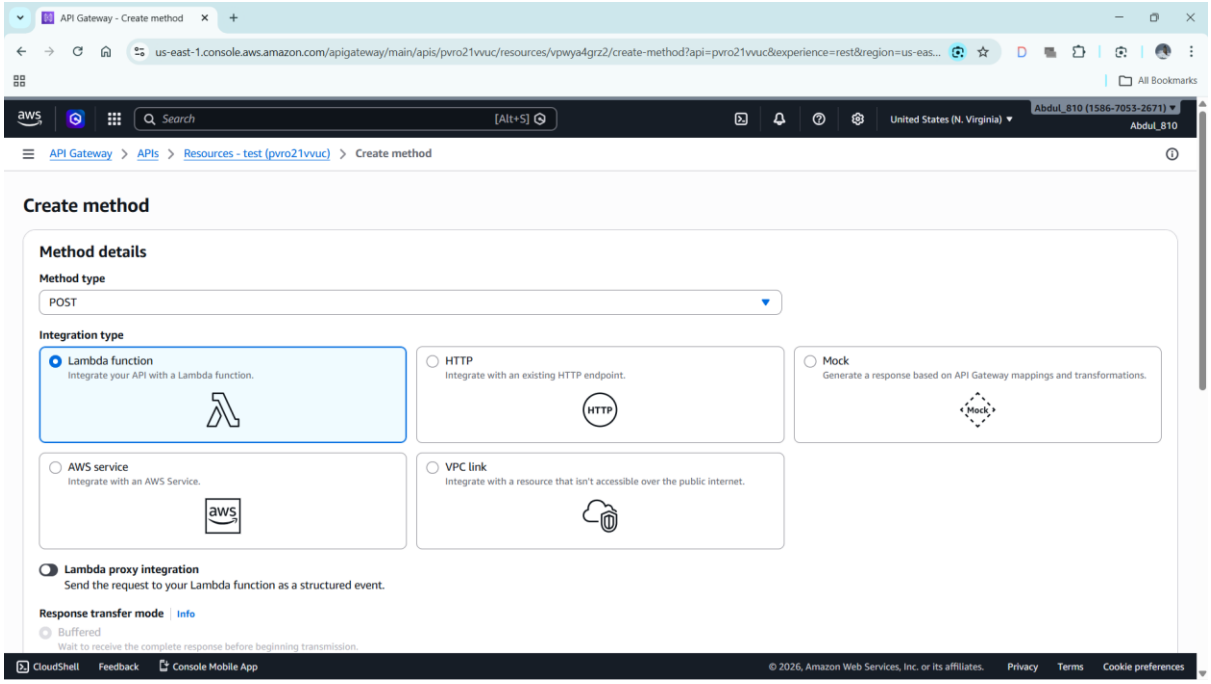


Name the API as “test” and click on Create API.

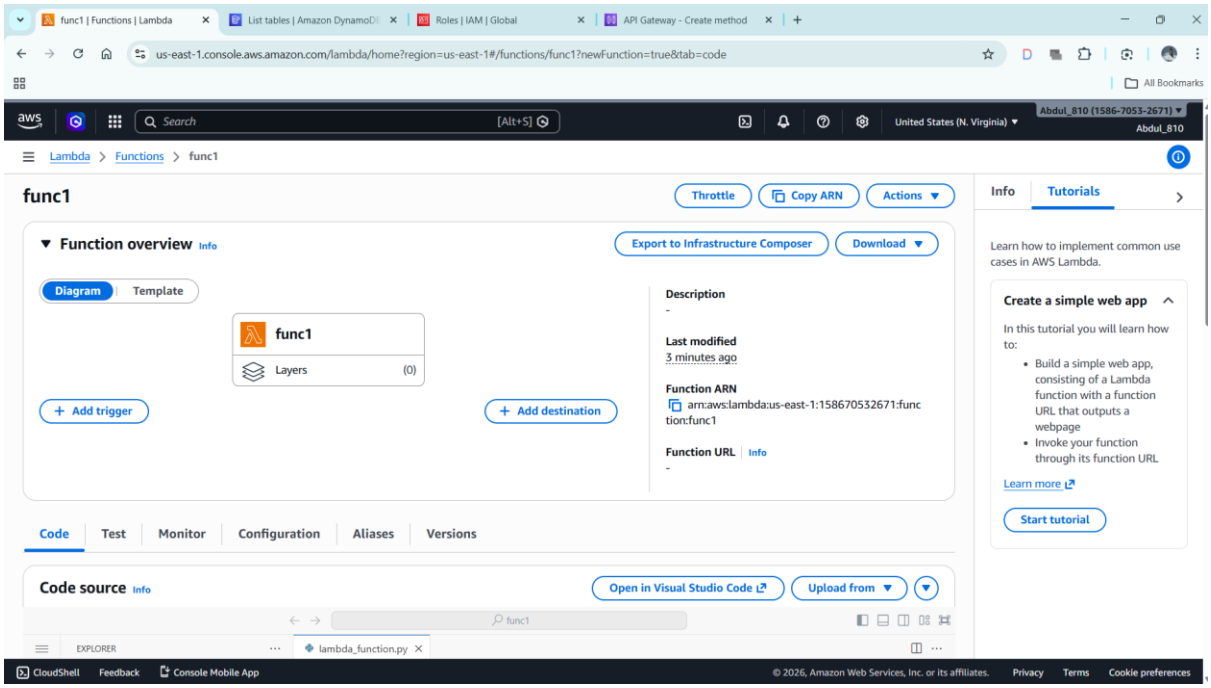
You’ll see that REST API is created successfully with the given name.



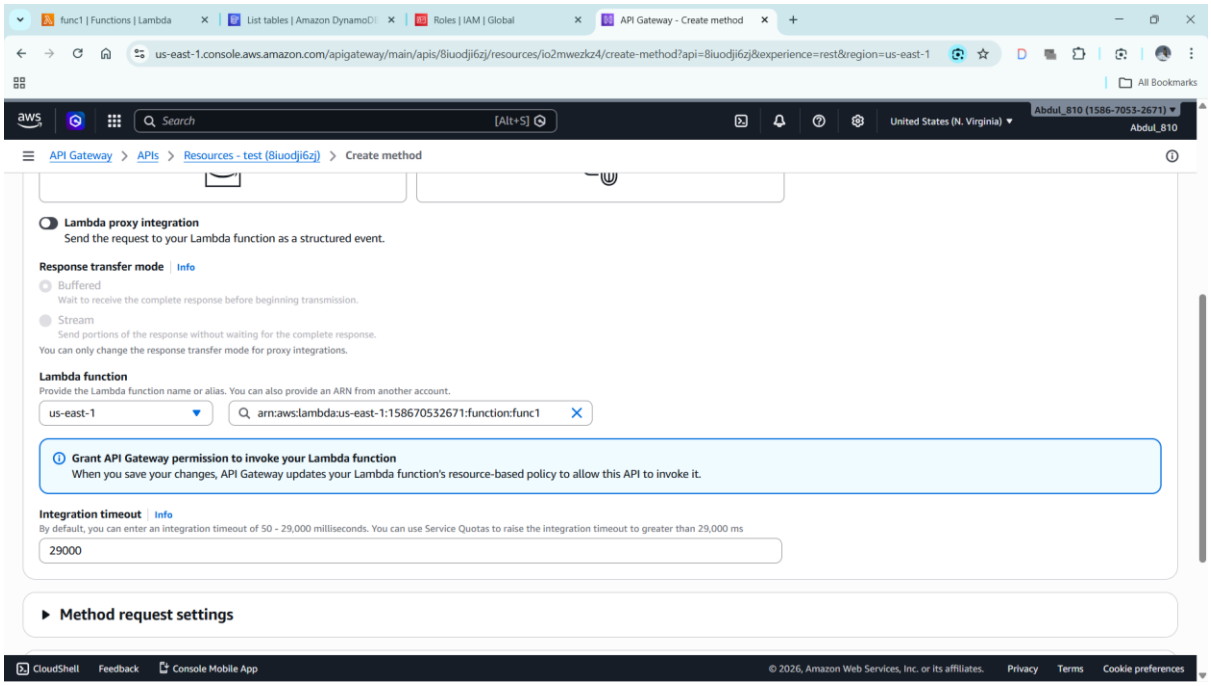
Later, click on create a method from the resources, and select method type as ‘POST’



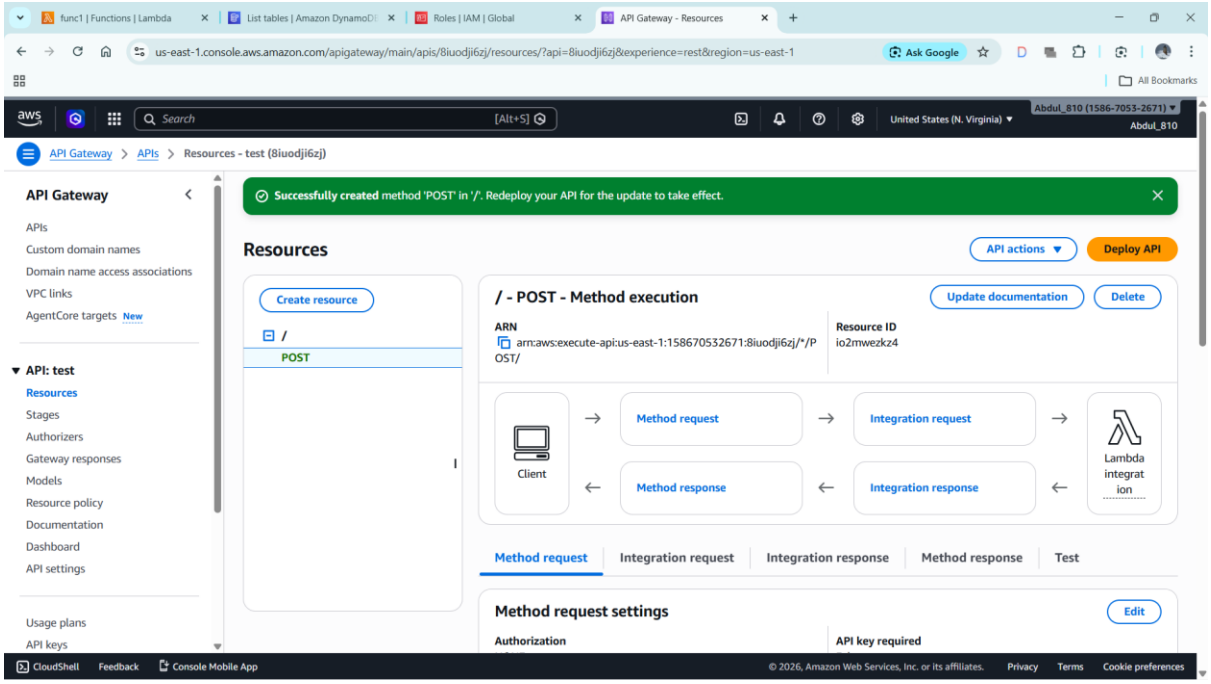
Copy the ARN from the lambda function which you have created earlier.



And paste it in the field where it has been asked



You'll see that method is created.

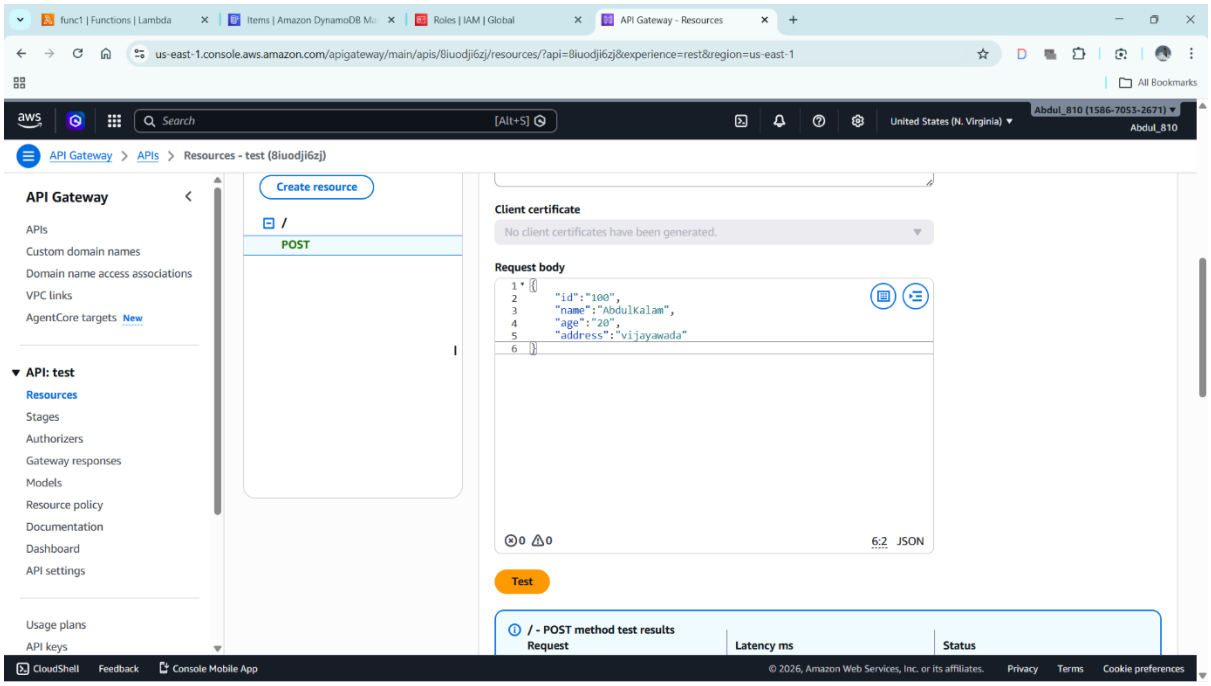
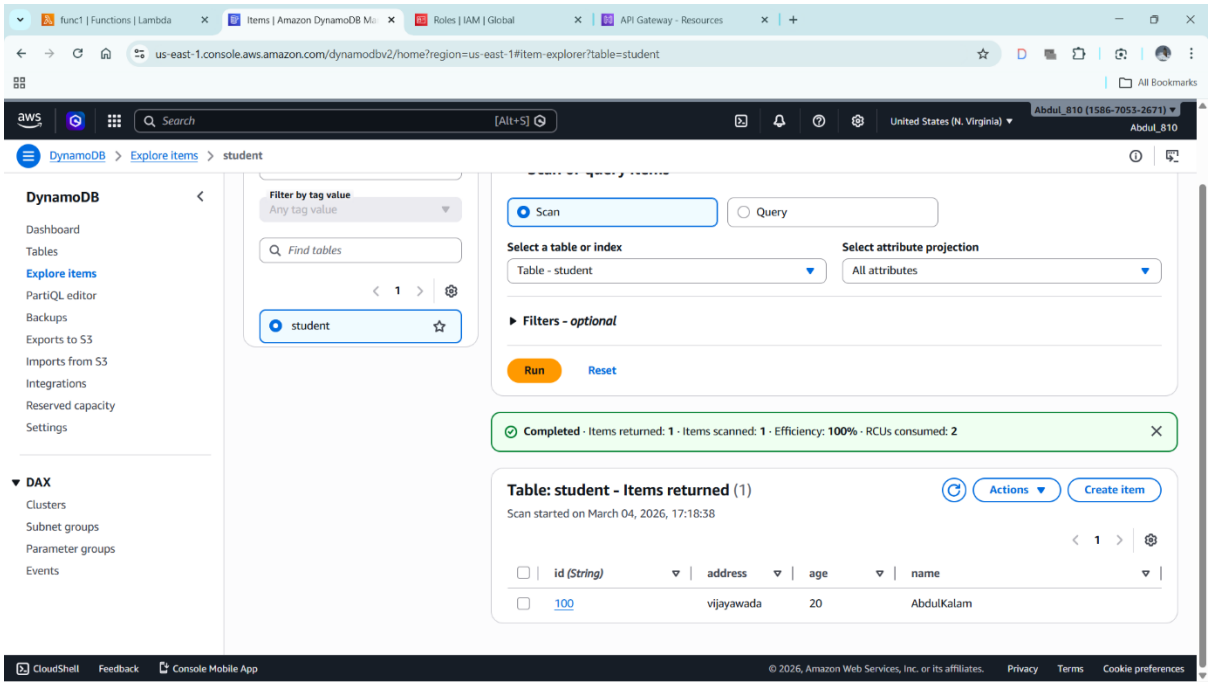


Now, click on test and scroll down to see 'request body' editor.

And paste the code given below:(This is JSON input data)

```
{
   "id":"100",
   "name": "Abdul Kalam",
   "age":"20",
   "address":"vijayawada"
}
```

And click on “test”.



Step-6: Now, Go to DynamoDB test table and explore the table items. The Data passed through API test will be available in the table(student) which was created earlier.